2.4 Introducing Polynomials

Lesson Introduction

Benchmark	MA.912.AR.1.3 Add, subtract, and multiply polynomial expressions with rational number coefficients.	Learning Targets	• I can write polynomials in standard form. • I can multiply a monomial to a polynomial.

Benchmark Coherence	Building On	Working Toward
	MA.8.AR.1.2 Apply properties of operations to multiply two linear expressions with rational coefficients.	MA.912.AR.1.5 Divide polynomial expressions using long division, synthetic division, or algebraic manipulation.

Essential Questions	• How do you write a polynomial in standard form? • How do you find the degree of a monomial? • How do you find the degree of a polynomial? • What determines which coefficient is the leading coefficient? • What property of exponents is used when multiplying a monomial to a polynomial?
Lesson Narrative	In this lesson, students will extend their prior work with using the product of powers law and multiplying monomials earlier in the unit to include multiplying a monomial by a polynomial. This includes a brief exploration connecting students' prior knowledge of multiplying using the distributive property as it applies to polynomials. The lesson moves deeper into introducing what polynomials are, how they are named, and how to write them in standard form. Students will be given multiple opportunities to make connections to prior knowledge as they build new understandings about polynomial expressions.

Component Summary				
	2.4.1 Warm-Up (~5 min.)	Completing a number puzzle *(SE p. 73)*	2.4.4 Collaboration (10-15 min.)	Identify the parts and features of a polynomial *(SE p. 76)*
	2.4.2 Exploration (10-15 min.)	Developing a definition for polynomials *(SE p. 73)*	2.4.5 Refresher (5 min.)	Reviewing the distributive property and how it applies to polynomial multiplication *(SE p. 77)*
	2.4.3 Guided Instruction (5-10 min.)	Identify degrees of monomials and polynomials, leading coefficients, and writing polynomials in standard form *(SE p. 75)*	2.4.6 Guided Practice (10-15 min.)	Multiplying monomials by polynomials *(SE p. 78)*
	2.4.7 Wrap-Up (~5 min.)	Growth-mindset reflection on mistakes made *(SE p. 79)*		

Resources	Glossary	Materials	Additional Preparation
	Key Terms: • Monomial • Degree of a monomial • Polynomial • Degree of a polynomial • Leading Coefficient Additional Terms: • Standard form • Distributive property of multiplication over addition	• (Optional) Graphing Utility • (Optional) Colored Pencils or Highlighters • (Optional) Scissors	• N/A

<table>
<tr><td rowspan="4">Teacher Tips</td><th>Common Misconceptions</th><th>Things to Consider</th></tr>
<tr><td>Students may make sign errors when multiplying.
Students may think the monomial to be distributed must always sit in front of the polynomial.
Students may incorrectly apply exponent laws when distributing a monomial to a polynomial.</td><td>2.4.1 Warm-Up is meant to begin the lesson with an engaging task. It may be easy to get stuck here, so it will be helpful to use a timer.
Give students the opportunity to generalize rules about monomials, determine the meaning of the prefixes used with polynomials, and make sense of standard form to help build a stronger foundation for understanding moving forward in the remainder of the unit.</td></tr>
<tr><th>Literacy and Language Routines</th><th>Mathematical Thinking and Reasoning (MTR) Standard</th></tr>
<tr><td>Notice and Wonder
2.4.2 Exploration positions students to notice the features of polynomials by analyzing a list of examples and non-examples. Consider applying this routine on both lists (examples and non-examples), encouraging students to discover the features of polynomials.</td><td>MA.K12.MTR.2.1: Multiple Representations
Students will be working to create a written definition for polynomials and write example polynomials in the 2.4.2 Exploration. Clarifications within this standard emphasize connections between concept and representation. Utilize the progression of the lesson to highlight the process by which students are moving from their written definition to writing example polynomials that involve specific features.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">Throughout the lesson, students will work with partners to enrich discussion about the content and give multiple students an entry-point into the content. During the 2.4.2 Exploration and 2.4.4 Collaboration, consider partnering students strategically to ensure rich discussions among students.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
</table>

2.4.1 Warm-Up: Number Puzzle

Component Overview: *Students solve a number puzzle where the sum of the rows and columns match the given numbers.*

2.4.1 Warm-Up: Number Puzzle

1. Solve the puzzle by filling in each blank cell in the blue box so that each row and column has the sum shown. Each cell must contain a different value.

   Answers will vary.

75	25	100
30	90	120
105	115	

While students are working independently, monitor and note students for a variety of answers to the puzzle. Probe these students to share their thought process during the whole-group discussion.

2.4.2 Exploration: Defining a Polynomial

Component Overview: *Students will explore the definition of a polynomial. Monitor students as they work to discover the types and features of polynomials. Students will not know the formal definition of a polynomial but should be encouraged to write a definition based on the features they listed. After students refine their definitions, provide the formal definition of a polynomial.*

2.4.2 Exploration: Defining a Polynomial

1. The charts show several examples and non-examples of polynomials.

Polynomials

$r^5 + 2.7r^3 - 0.3r$

$\frac{3}{4}x^2y$ $\quad 80z^{15}$

$b - \frac{1}{5}$ $\quad 58c^0$ $\quad \frac{24k}{3}$

$\sqrt{9}np$ $\quad (m - 16)$

Not Polynomials

$r^5 + 2.7r^{-3} - 0.3r$

$\frac{3x^2}{4y}$ $\quad 80z^{1.5}$

$\frac{1}{b-5}$ $\quad 58c^x$ $\quad 24k^{\frac{1}{2}}$

$\sqrt{9np}$ $\quad (m - 16)^{-1}$

a. Create a list of identifying features polynomials can have versus those polynomials cannot have.

Polynomials can have …	Polynomials cannot have …
Answers will vary	*Answers will vary*
Fractional coefficients	*Negative exponents*
Exponents of 0	*Fractional exponents*
Square root coefficients	*Square root variables*

b. Compare your lists with your partner. Add any additional features you found after your discussion.
Answers will vary.

c. With your partner, write a definition for polynomial.
Answers will vary. A polynomial is a collection of terms that have whole number exponents and no variables in a denominator. A polynomial is a series of terms that contain exponents that are not negative, do not appear in the denominator, and are not radical.

Consider using this routine while students analyze the examples and non-examples of polynomials. Encourage students to take note of patterns and trends they notice between the types of numbers being used in the examples and non-examples.

If students struggle to write any features, consider asking the following.

- *What is different about the polynomials in the polynomial box from the expressions in the corresponding location in the not polynomial box?*
- *Is there a term from either list that stands out to you? Why?*
- *What do you notice about the location of any fractions you see?*
- *What do you notice about the signs of the exponents?*

Consider providing students with sentence stems to support rich discussion when comparing their lists.

- *Polynomials can have ______ because …*
- *Polynomials cannot have ______ because …*
- *I agree with you because …*
- *I disagree with you because …*
- *One thing I noticed was …*

d. Ask two other pairs of partners the definition they wrote. Listen to their definition and write a summary. *You are the only person who should write in the first column of the table.* Have the person initial next to your summary stating that your summary of their definition was accurate.

My Summary of the Definition of a Polynomial	Initials
Answers will vary.	

e. After reviewing the definitions created by your classmates, go back to your definition and make revisions, if needed.
Answers will vary.

Some polynomials are described with specific names based on the number of terms.

2. Four prefix options are given.

Prefixes			
tri-	mono-	poly-	bi-

a. Match the prefixes to identify the types of polynomials in the table based on the description and examples.

Polynomials		
monomials (one term)	binomials (two terms)	trinomials (three terms)
$-7.8t^3w$	$-7.8t^3w + 2.04w$	$-7.8t^3w + 2.04w - t$
14	$-\frac{8}{9}y^2 + 14$	$-\frac{8}{9}y^2 - 12m + 14$
$9x$	$x + 2$	$5x^2 + 2x + 4$

b. In the last row of the table, create your own polynomial that fits each category.
Answers will vary. Sample answers in table above.

Challenge students to create an example and non-example for the definitions provided by their classmates. Position students to try to find "loopholes" in the definitions provided and suggest ways to refine their summaries.

Students are not just copying their classmates' observations, but instead critically listening and taking notes. Students record their classmates' observations for additional entry points and scaffold when making sense of the observations made.

After students complete the table, consider asking the following.

- *Which prefix was not used in the table?*
- *How would a polynomial with that prefix look? How do you know?*
- *How do you intend on remembering which prefix goes with each number of terms?*

2.4.3 Guided Instruction: Polynomials and Standard Form

Component Overview: *Students will be formally introduced to polynomials and standard form. Use think-aloud strategies while modeling for students throughout this component of the lesson. Explain how the answers are determined for students to better understand the logic behind each answer.*

2.4.3 Guided Instruction: Polynomials and Standard Form

A ***monomial*** is a polynomial with one term.

The **degree of a monomial** is the sum of the exponents of the variables in a monomial.

1. Identify the degree of each monomial.

Monomial	Degree
$-4x^5y^6$	11
$1.6p^6c^5w^8$	19
$-\frac{7}{11}rh^{24}$	25

A ***polynomial*** is the sum or difference of terms which have variables raised to non-negative integer powers and which have coefficients that may be real or complex.

The **degree of a polynomial** is the greatest degree of the monomials in a polynomial.

2. What is the degree of $62pk^8 + 7a^2b^5 - 38g^4h^6$?
10

Consider using different colors, circles, or underlining different parts of the polynomials to provide both visual and auditory entry points. For example, in question 2, circle all the exponents to position students to see that the exponents are the focus when finding the degree of the polynomial. Then, continue by finding the degree of each monomial to determine the degree of the polynomial.

Specifically have students make connections to their responses to the first part of this question. Students who may not have made the connection to the importance of order might now better connect when given a real-world context.

Consider a Foldable, Frayer Model, or type of interactive notebook for students to summarize new information from this component. Examples, non-examples, and verbal descriptions will provide multiple pathways for students to engage with the content in meaningful ways that encourage conceptual understanding over rote memorization.

The **leading coefficient** is the coefficient of the first term of a polynomial whose terms are written in descending order from largest degree to smallest degree.

3. What is the leading coefficient of $62pk^8 + 7a^2b^5 - 38g^4h^6$?

   -38

A polynomial in **standard form** is one where the term with the highest degree is written first and the other terms are in descending order by degree. The variables in each term should be written in alphabetical order.

4. Rewrite polynomial $62pk^8 + 7a^2b^5 - 38g^4h^6$ in standard form.

   $-38g^4h^6 + 62k^8p + 7a^2b^5$

2.4.4 Collaboration: Polynomials and Standard Form

Component Overview: *Students will collaborate to determine the basic information for given polynomials. Monitor students as they work in pairs to apply the knowledge gained from 2.4.3 Guided Instruction.*

2.4.4 Collaboration: Polynomials and Standard Form

Work with your partner to complete the following.

1. Three polynomials are given in the table.

   a. Determine whether each polynomial is written in standard form. If it is not, rewrite it in standard form.

   b. Then, in the second column, identify the features of each polynomial.

$-52a + 3a^3b^4 + 21a^2b^2$ $3a^3b^4 + 21a^2b^2 - 52a$	Number of terms ___3___ Degree of the polynomial ___7___ Leading term ___$3a^3b^4$___ Leading coefficient ___3___
$-9x^8y^4 + 2x^9y^2$	Number of terms ___2___ Degree of the polynomial ___12___ Leading term ___$-9x^8y^4$___ Leading coefficient ___-9___
$5m^6 + 13m^2 - \frac{1}{4}m^8$ $-\frac{1}{4}m^8 + 5m^6 + 13m^2$	Number of terms ___3___ Degree of the polynomial ___8___ Leading term ___$-\frac{1}{4}m^8$___ Leading coefficient ___$-\frac{1}{4}$___

If pairs finish early, challenge the students to write an additional polynomial for each row that matches the first two parts listed in the 2nd column. For a row one example, students may write a 3-term polynomial with a degree of 7, such as: $-2x^5y^2 + 7x^2y - 6$.

2.4.4 Collaboration: Polynomials and Standard Form (continued)

2. With your partner, match the polynomial in the left column by drawing an arrow to its descriptive feature in the right column.

$x^3 + 4x^2 - 5x + 9$
$45a^2b^3$
$3x^4 - 9x^3 + 4x^9$
$7a^6b^2 + 18ab^3c - 9a^7$
$x^5 - 9x^3 + 2x^7$
$3x^3 + 7x^2 - 11$
$x^2 - 2$

fifth-degree polynomial
constant term of -2
seventh-degree polynomial
leading coefficient of 3
four terms
eighth-degree polynomial
equivalent to $4x^9 + 3x^4 - 9x^3$

Consider adjusting the matching activity by transitioning it using the Information Gap routine to encourage rich discussion between students. Cut the table into pieces and split the pieces between the two students. Students then take turns describing one of their cards without showing their card to their partner while trying to find the correct matches.

2.4.5 Refresher: Distributing a Monomial to a Polynomial

Component Overview: *Students complete two questions with their partners prior to a whole-group discussion to ensure collective understanding.*

2.4.5 Refresher: Distributing a Monomial to a Polynomial

The ***distributive property of multiplication over addition*** applies to polynomials. Recall that the distributive property "distributes" a factor to all terms within a set of parentheses.

1. An example of distributing a monomial to a polynomial is shown.

$$-3b^2\left(\frac{1}{4}m^3 + 2m - 8\right)$$

$$-\frac{3}{4}b^2m^3 - 6b^2m + 24b^2$$

   a. Discuss with your partner what operation the arrows are demonstrating in the example.
   The arrows represent multiplication.

   b. In the resulting product, why are the first two terms negative while the third term is positive?
   Multiplying the negative monomial on the outside of the parenthesis to the first two positive monomials inside the parenthesis results in two negative terms. Then, multiplying the negative monomial to the negative third term results in a positive third term.

After question 1a, consider asking the following.

- *What exponent law is being applied when distributing a monomial to a polynomial?*

2.4.6 Guided Practice: Distributing a Monomial to a Polynomial

Component Overview: *Students will practice applying the distributive property from a monomial to a polynomial. Monitor students as they work to multiply monomials to polynomials. Use questioning to help students find entry points when needed.*

2.4.6 Guided Practice: Distributing a Monomial to a Polynomial

1. Find each product by distributing the monomial to the polynomial. Then, complete the table for the resulting product.

$(1 + 3y^3 - 2y^4)(-6y^4)$	
$-6y^4 - 18y^7 + 12y^8$	
Product in Standard Form $12y^8 - 18y^7 - 6y^4$	
Leading Coefficient 12	Degree of Polynomial 8

$-6dg^4(d^9g^2 - 2d^4g^2 + 10)$	
$-6d^{10}g^6 + 12d^5g^6 - 60dg^4$	
Product in Standard Form $-6d^{10}g^6 + 12d^5g^6 - 60dg^4$	
Leading Coefficient -6	Degree of Polynomial 16

Consider asking the following when reviewing these questions.

- *How did you determine the exponents and coefficients after distributing the monomial?*
- *What mathematical operations were performed?*
- *When finding the degree and leading coefficient, do you find it easier to write the polynomial in standard form first? Why or why not?*
- *What exponent law was used?*

For students who struggle with distributing when negative terms are involved, it is helpful to have them create a quick sign chart to identify the sign the resulting product will be.

2.4.6 Guided Practice: Distributing a Monomial to a Polynomial (continued)

$\frac{3}{2}a^2\left(8a - 9b^2 + \frac{1}{3}a^3b\right)$	
$12a^3 - \frac{27}{2}a^2b^2 + \frac{1}{2}a^5b$	
Product in Standard Form $\frac{1}{2}a^5b - \frac{27}{2}a^2b^2 + 12a^3$	
Leading Coefficient $\frac{1}{2}$	Degree of Polynomial 6
$-0.25b^3c(1.5c + 6b^2 - 4b^5c^3)$	
$-0.375b^3c^2 - 1.5b^5c + b^8c^4$	
Product in Standard Form $b^8c^4 - 1.5b^5c - 0.375b^3c^2$	
Leading Coefficient 1	Degree of Polynomial 12

Consider providing students with choice by allowing them to choose 3 of the 4 problems to complete and show their understanding of monomial and polynomial multiplication.

2.4.7 Wrap-Up: Growth Mindset

Component Overview: *Students will assess their own actions independently and provide feedback regarding their mistakes during today's lesson.*

2.4.7 Wrap-Up: Growth Mindset

1. Mistakes are an opportunity to learn. Complete the statement to highlight a mistake you learned from.

   My favorite mistake from today was …

Means
I
Start
To
Acquire
Knowledge
Experience
Skills

Answers will vary.

After students identify their favorite mistake, challenge students to create an Error Analysis problem using that mistake for a classmate to solve.